



CHAPTER

Diluate Tank

Buffers the desalinated stream through recirculation, on its way to the customer

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15.1 Overview and function

The diluate tank (tag prefix **TD**) buffers the desalinated stream as it recirculates through the SonixED™ stack(s) (chapter 10) — diluate is not necessarily delivered in a single pass; it recirculates through this tank and back through the stack until the target product conductivity is reached, at which point it passes on to the product/delivery side of the system (tag prefix **TB**, for “breaking” out of the recirculation loop) for delivery to the customer. This is the tank whose reading is the unit’s headline water- quality result: the HMI Overview screen’s “product conductivity” KPI (chapter 6) is read from this stream.



Figure 15.1. Suggested figure: photograph of the diluate tank(s), labeled with their connections (from-stack, to-stack recirculation, to-product-tank outlet) and instrumentation.

15.2 Key specifications

No tank capacity/material specification exists yet for a finalized standard SalinBloc™ bill of materials, but two real, recent tank orders placed by HydroVolta give a concrete sense of the sizes used: a Rotterdam Plastics order (Oct. 2025) for one 800-litre and one 450-litre vertical polyethylene tank, and a Boralit/Galo Tanks order for two 1000-litre rectangular polyethylene tanks (RT1000RH ZW series). TODO: which of these is the diluate tank was not confirmed in the order documents — do not assume a mapping without checking the as-built P&ID for the specific unit. Some documented configurations use **two** diluate buffer tanks in series (tagged **BTD1/BTD2**, distinct from the main TD tank) rather than one — confirm which arrangement applies to the unit at hand.

- Level: TD_Level (main tank); BTD1/BTD2 level alarms exist in some configurations.
- Process instrumentation on this stream: pH (TD_pH), pressure (TD_pressure), flow (TD_Flow) — see chapter 13 for sensor makes/models.
- Downstream product tank (TB): conductivity (TB_Cond) and level (TB_Level) — this is the KROHNE SMARTPAT COND reading behind the HMI's "product conductivity" KPI (chapters 6 and 13).

15.3 Operation

Diluate recirculates through this tank and the stack under automatic control during **Production** mode (chapter 6); level is monitored continuously with high/low alarms confirmed at **40 cm (low)** / **90 cm (high)** in a documented configuration (see the warning-code table in appendix A) — TODO confirm these setpoints for the standard product's tank sizing. Product only passes to the TB (delivery) tank once conductivity is within its target range.

15.4 Maintenance

TODO: no tank cleaning/inspection schedule was found. Because this tank holds treated (low-salinity) water rather than raw feed or concentrate, the main risk during idle periods is biological growth (algae/biofilm) rather than scaling — inspect at the interval in chapter 7 once one is confirmed.

15.5 Troubleshooting

HL_Alarm_TD/LL_Alarm_TD cover the main tank (plus equivalent alarms for BTD1/BTD2 where fitted). On any level alarm, check the relevant level sensor per the bench-test procedure in section 13.4 before assuming a process fault. If product conductivity (TB) is out of range while the tank level itself is normal, the fault is more likely in the stack's reversal timing or feedwater quality (chapter 10) than in the tank itself.